



BEST PRACTICE SERIES

**Improving Availability through
Dealer/Customer PM Partnering**

Application Maintenance **Site Management** Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

Significant improvements in machine availability can be gained through an integrated partnership between dealer and customer mechanics for the preventative maintenance process. Availability is improved by increasing the PM process efficiency and better completion of backlog repairs.

Low availability and frequent unscheduled repairs drove a restructuring of the PM process. A full time dealer mechanic dedicated to PM's was put in place to:

1. Assist with minor mechanic repairs,
2. Ensure complete inspections,
3. Create/manage backlogs to be performed in the repair shop immediately after PM completion.

The number of site mechanics was also increased to enable the completion of as many minor repairs as possible while minimizing the time spent in PM.

Dealer mechanic assistance, sufficient manpower, disciplined execution of procedures, and accurate backlog management resulted in increased machine availability and decreased unscheduled shop repair time.

2.0 Best Practice Description

The duration of preventative maintenance for each machine is four hours.

- 1 hour in wash bay
- 1 hour for lubrication
- 2 hours of mechanical repair

Lubrication activities include

- Oil replacement,
- Kidney loop filtration for remaining oil,
- Collecting of SOS samples,
- Cutting and examination of every filter.

The mechanical repair bay work is driven by a serial number specific clipboard documenting the activities to be completed.

Clipboard includes:

- Backlog repair work orders
- Mag plug inspection sheet
- Operator "fix list"
- Checklist of inspections and functionality checks

The dealer mechanic handles the VIMS download and acts upon the data to perform additional diagnostics or repairs.

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Overall, the dealer mechanic resides in the mechanical repair bay to complete as many backlogs as possible and ensure that the remaining are documented accurately and addressed thereafter in the main repair shop. The main repair shop will keep the machine as long as it takes to complete all backlogs unless lacking parts or resources force the work to be postponed until next PM.

On a daily basis, one PM mechanic works the fuel island to check lube levels, perform general inspections, check for leaks, and document operator feedback.

3.0 Implementation Steps

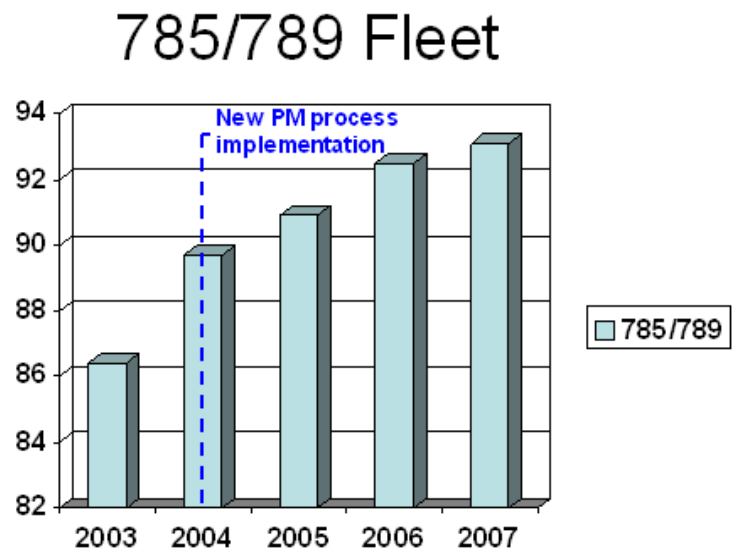
- Identify and train PM customer and dealer mechanics.
- Provide adequate separate areas for wash, lube, and mechanical repair.
- Provide inspection checklists for filters, mag plugs, structures, and functional checks.
- Provide kidney lube filtration equipment and efficient means to cut filters and document findings for further investigation.
- Create management system for backlog, work order, and inspection documentation to be available to the mechanics by serial number.
- Provide all required PM tools.
- Provide means for the dealer mechanic to provide accurate and timely communication of backlog repairs to the main shop.

4.0 Benefits

Increased availability and decrease in unscheduled repairs.

Truck availability improvement graph:

- Dealer and site started the new PM process in mid-2006 among other operations changes:
- **Main shop is running 72% scheduled work for trucks and 80 – 90% for all other machines combined.**



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5.0 Resources Required

1. Target eight mechanics per shift broken down as follows:
(including the one dealer mechanic in the mechanical bay)
 - Wash bay: 2
 - Lube bay: 3 – 4
 - Mechanical bay: 2 – 3
2. Kidney loop filter cart
3. Filter cutting and inspection bench
4. System to communication work order and checklist information

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

- [0806-2.7-1008](#) - Backlog Management
- [1006-2.13-1027](#) - Utilizing VIMS Data for Effective Planned Maintenance
- [0806-2.7-1007](#) - Condition Monitoring – Field Inspection
- [1106-2.14-1036](#) - Oil Filter Cutting Bench

8.0 Acknowledgements

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